



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Unit-4 Activity

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7/826

1. A university wants to determine whether students who spend more hours studying obtain higher marks.

Dataset

StudyHours	Marks
2	45
3	52
4	58
5	63
6	69
7	74
8	81
9	88

Task

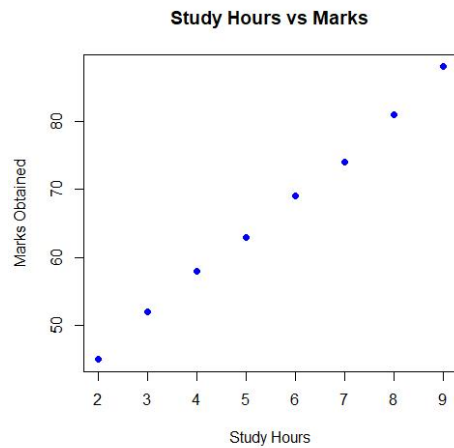
- **Create the dataset in R.**
- **Draw a scatterplot.**
- **Add axis labels and title.**
- **Interpret whether a positive correlation exists.**

Code:

```
StudyHours <- c(2, 3, 4, 5, 6, 7, 8, 9)
Marks <- c(45, 52, 58, 63, 69, 74, 81, 88)
student_data <- data.frame(StudyHours, Marks)
print(student_data)
plot(student_data$StudyHours,
      student_data$Marks,
      main = "Study Hours vs Marks",
      xlab = "Study Hours",
      ylab = "Marks Obtained",
      pch = 19,
      col = "blue")
```

```
abline(lm(Marks ~ StudyHours, data = student_data),
      col = "red",
      lwd = 2)
```

Output:

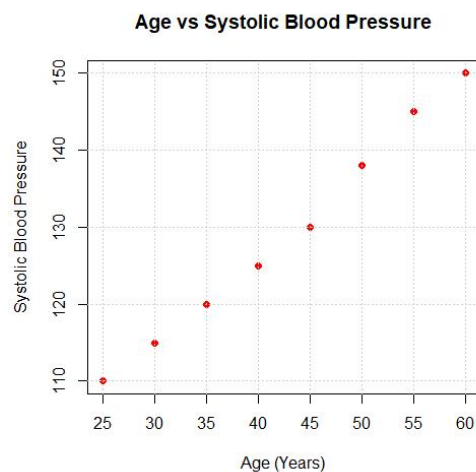


2. A hospital records patients' age and systolic blood pressure. Write a Python program using Matplotlib to: Create a scatterplot. Display gridlines. Change point color to red. Comment on the observed trend.

Code:

```
age <- c(25, 30, 35, 40, 45, 50, 55, 60)
bp <- c(110, 115, 120, 125, 130, 138, 145, 150)
plot(age, bp,
     main = "Age vs Systolic Blood Pressure",
     xlab = "Age (Years)",
     ylab = "Systolic Blood Pressure",
     col = "red",
     pch = 19)
grid()
```

Output:

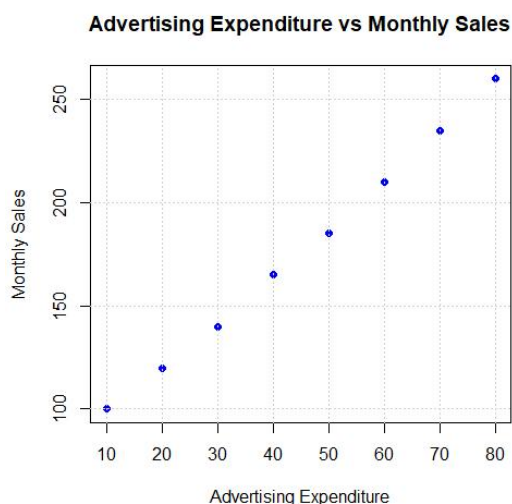


3. A company wants to investigate whether advertising expenditure influences monthly sales. Develop a scatterplot and determine whether sales increase with advertising expenditure. A financial institution has collected information on: Income Savings Loan Amount Credit Score Generate a correlation matrix and visualize it using a correlogram. Which variables exhibit the strongest positive correlation?

Code:

```
advertising <- c(10, 20, 30, 40, 50, 60, 70, 80)
sales <- c(100, 120, 140, 165, 185, 210, 235, 260)
plot(advertising, sales,
     main = "Advertising Expenditure vs Monthly Sales",
     xlab = "Advertising Expenditure",
     ylab = "Monthly Sales",
     col = "blue",
     pch = 19)
grid()
```

Output:



3. A financial institution has collected information on: Income Savings Loan Amount Credit Score Generate a correlation matrix and visualize it using a correlogram. Which variables exhibit the strongest positive correlation?

Code:

```
install.packages("corrplot")
library(corrplot)
finance <- data.frame(
  Income = c(30000, 40000, 50000, 60000, 70000, 80000, 90000, 100000),
  Savings = c(5000, 8000, 12000, 18000, 24000, 30000, 38000, 45000),
  LoanAmount = c(15000, 20000, 28000, 35000, 42000, 50000, 60000, 70000),
  CreditScore = c(620, 650, 680, 710, 740, 770, 800, 830)
)
cor_matrix <- cor(finance)
print(cor_matrix)
```

```
corrplot(cor_matrix,
         method = "color",
         type = "upper",
         addCoef.col = "black",
         tl.col = "black")
```

Output:



5. Using the Iris dataset,.Compute the correlation matrix.Display it as a heatmap using Seaborn.Identify variables having negative correlation.

Code:

```
install.packages("corrplot")
library(corrplot)
data(iris)
iris_data <- iris[, 1:4]
cor_matrix <- cor(iris_data)
print(cor_matrix)
corrplot(cor_matrix,
         method = "color",
         type = "upper",
         addCoef.col = "black",
         tl.col = "black")
```

Output:

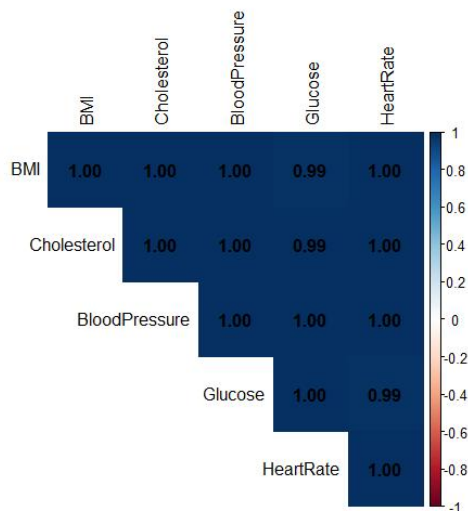


6. A healthcare dataset contains: BMI Cholesterol Blood Pressure Glucose Heart Rate Create a correlogram and identify multicollinearity among variables.

Code:

```
install.packages("corrplot")
library(corrplot)
health <- data.frame(
  BMI = c(22, 24, 26, 28, 30, 32, 34, 36),
  Cholesterol = c(180, 190, 200, 210, 220, 235, 245, 260),
  BloodPressure = c(110, 115, 120, 125, 130, 138, 145, 150),
  Glucose = c(85, 90, 95, 100, 110, 120, 130, 140),
  HeartRate = c(68, 70, 72, 74, 76, 78, 80, 82)
)
print(health)
cor_matrix <- cor(health)
print(cor_matrix)
corrplot(cor_matrix,
  method = "color",
  type = "upper",
  tl.col = "black",
  addCoef.col = "black")
cat("\nVariables with correlation greater than 0.90 indicate multicollinearity.\n")
```

Output:

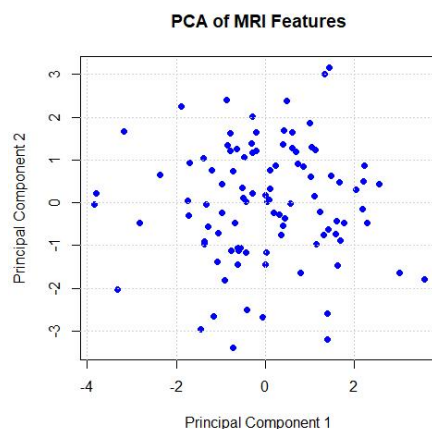


7. A researcher is analyzing 25 features extracted from MRI images. Perform PCA to reduce the dimensionality to two principal components. Plot the transformed data.

Code:

```
set.seed(123)
mri <- data.frame(matrix(runif(100 * 25), nrow = 100, ncol = 25))
colnames(mri) <- paste0("Feature", 1:25)
pca_result <- prcomp(mri, scale. = TRUE)
summary(pca_result)
pca_data <- data.frame(PC1 = pca_result$x[,1],
                       PC2 = pca_result$x[,2])
plot(pca_data$PC1,
     pca_data$PC2,
     col = "blue",
     pch = 19,
     xlab = "Principal Component 1",
     ylab = "Principal Component 2",
     main = "PCA of MRI Features")
grid()
```

Output:

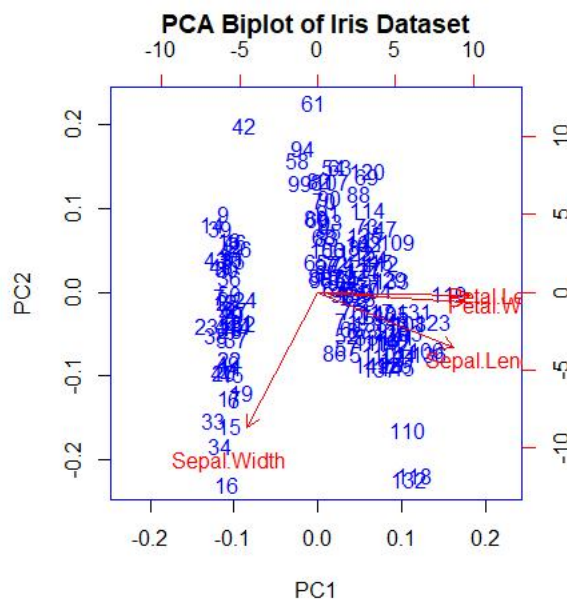


8. Using the Iris dataset, Perform PCA. Display the summary. Draw a biplot. Explain which principal component explains the highest variance.

Code:

```
data(iris)
iris_data <- iris[, 1:4]
pca_result <- prcomp(iris_data, scale. = TRUE)
summary(pca_result)
biplot(pca_result,
       main = "PCA Biplot of Iris Dataset",
       col = c("blue", "red"))
```

Output:



9. An automobile company has recorded 15 specifications for various cars. Reduce the dimensions using PCA and visualize the first two principal components.

Code:

```
set.seed(123)
cars_data <- data.frame(matrix(runif(100 * 15), nrow = 100, ncol = 15))

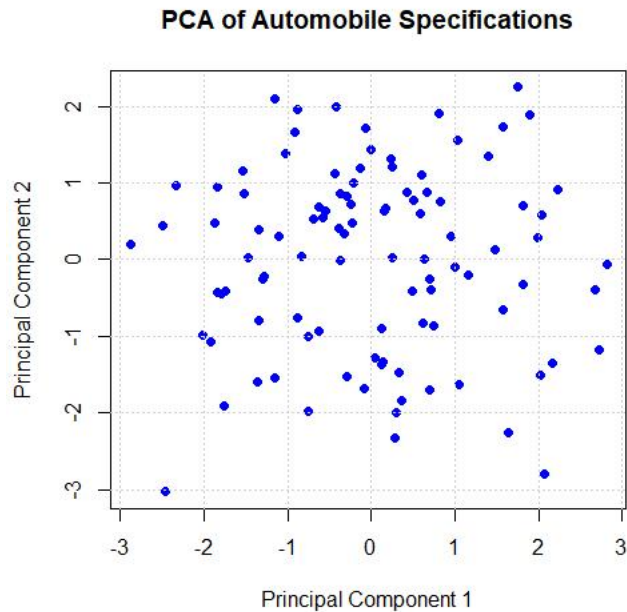
colnames(cars_data) <- paste0("Spec", 1:15)
pca_car <- prcomp(cars_data, scale. = TRUE)
summary(pca_car)
pca_df <- data.frame(PC1 = pca_car$x[,1],
                    PC2 = pca_car$x[,2])
plot(pca_df$PC1,
     pca_df$PC2,
     pch = 19,
     col = "blue",
     xlab = "Principal Component 1",
```

```

ylab = "Principal Component 2",
main = "PCA of Automobile Specifications")
grid()

```

Output:



10. A facial recognition dataset contains 128-dimensional facial embeddings. Use t-SNE to visualize the data in two dimensions. Explain why t-SNE is preferred over PCA for visualization.

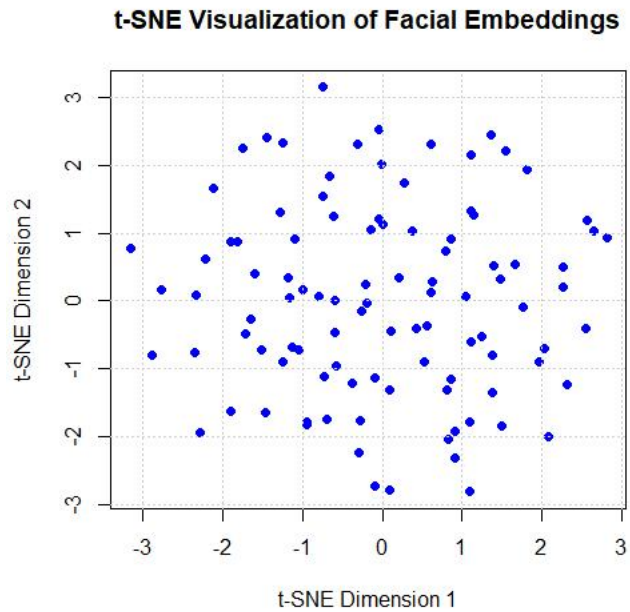
Code:

```

install.packages("Rtsne")
library(Rtsne)
set.seed(123)
face_data <- matrix(runif(100 * 128), nrow = 100, ncol = 128)
tsne_result <- Rtsne(face_data,
  dims = 2,
  perplexity = 30,
  verbose = TRUE,
  max_iter = 500)
plot(tsne_result$Y,
  col = "blue",
  pch = 19,
  xlab = "t-SNE Dimension 1",
  ylab = "t-SNE Dimension 2",
  main = "t-SNE Visualization of Facial Embeddings")
grid()

```

Output:

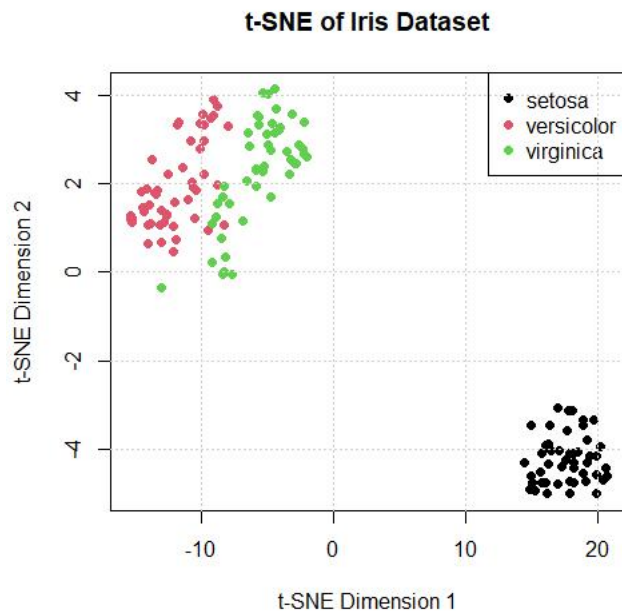


11. Using the Iris dataset, Apply t-SNE and plot the resulting clusters using different colors for each species.

Code:

```
install.packages("Rtsne")
library(Rtsne)
data(iris)
iris_features <- iris[, 1:4]
set.seed(123)
tsne_result <- Rtsne(as.matrix(iris_features),
                      dims = 2,
                      perplexity = 30,
plot(tsne_result$Y,
     col = as.numeric(iris$Species),
     pch = 19,
     xlab = "t-SNE Dimension 1",
     ylab = "t-SNE Dimension 2",
     main = "t-SNE of Iris Dataset")
legend("topright",
      legend = levels(iris$Species),
      col = 1:3,
      pch = 19)
grid()
```

Output:

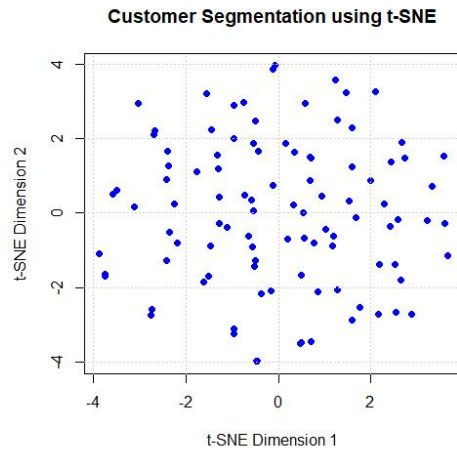


12. A customer segmentation dataset contains 40 behavioral variables. Visualize customer groups using t-SNE. Discuss whether distinct clusters are formed.

Code:

```
install.packages("Rtsne")
library(Rtsne)
set.seed(123)
customer_data <- matrix(runif(100 * 40), nrow = 100, ncol = 40)
tsne_result <- Rtsne(customer_data,
  dims = 2,
  perplexity = 30,
  check_duplicates = FALSE)
plot(tsne_result$Y,
  pch = 19,
  col = "blue",
  xlab = "t-SNE Dimension 1",
  ylab = "t-SNE Dimension 2",
  main = "Customer Segmentation using t-SNE")
grid()
```

Output:

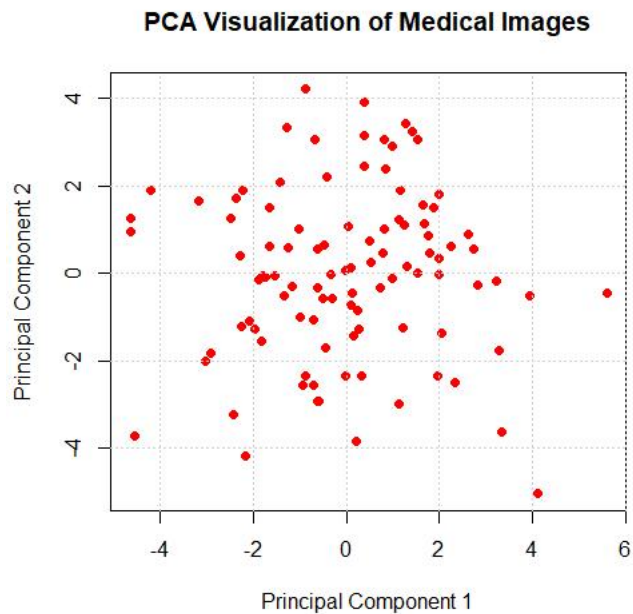


13. A medical image dataset contains 100 extracted features. Apply UMAP to reduce the dimensions to two. Visualize the output. Compare the result with PCA.

Code:

```
install.packages("umap")
library(umap)
set.seed(123)
medical_data <- matrix(runif(100 * 100), nrow = 100, ncol = 100)
umap_result <- umap(medical_data)
plot(umap_result$layout,
     pch = 19,
     col = "blue",
     xlab = "UMAP Dimension 1",
     ylab = "UMAP Dimension 2",
     main = "UMAP Visualization of Medical Images")
grid()
pca_result <- prcomp(medical_data, scale. = TRUE)
plot(pca_result$x[,1],
     pca_result$x[,2],
     pch = 19,
     col = "red",
     xlab = "Principal Component 1",
     ylab = "Principal Component 2",
     main = "PCA Visualization of Medical Images")
grid()
```

Output:

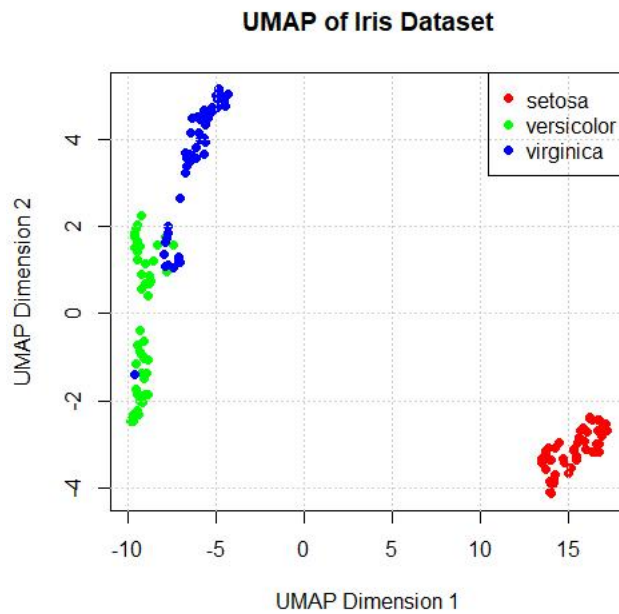


14. Apply UMAP to the Iris dataset. Plot the reduced dimensions using different colors for each species.

Code:

```
install.packages("umap")
library(umap)
data(iris)
iris_data <- iris[, 1:4]
set.seed(123)
umap_result <- umap(iris_data)
colors <- c("red", "green", "blue")
species_color <- colors[as.numeric(iris$Species)]
plot(umap_result$layout,
     col = species_color,
     pch = 19,
     xlab = "UMAP Dimension 1",
     ylab = "UMAP Dimension 2",
     main = "UMAP of Iris Dataset")
legend("topright",
     legend = levels(iris$Species),
     col = colors,
     pch = 19)
grid()
```

Output:



15. An e-commerce company has collected customer purchasing behavior consisting of 60 attributes.Reduce the dimensions using UMAP and identify customer clusters.

Code:

```
install.packages("umap")
library(umap)
set.seed(123)
customer_data <- matrix(runif(100 * 60), nrow = 100, ncol = 60)
umap_result <- umap(customer_data)
plot(umap_result$layout,
     pch = 19,
     col = "blue",
     xlab = "UMAP Dimension 1",
     ylab = "UMAP Dimension 2",
     main = "Customer Clusters using UMAP")
grid()
```

Output:

Customer Clusters using UMAP

